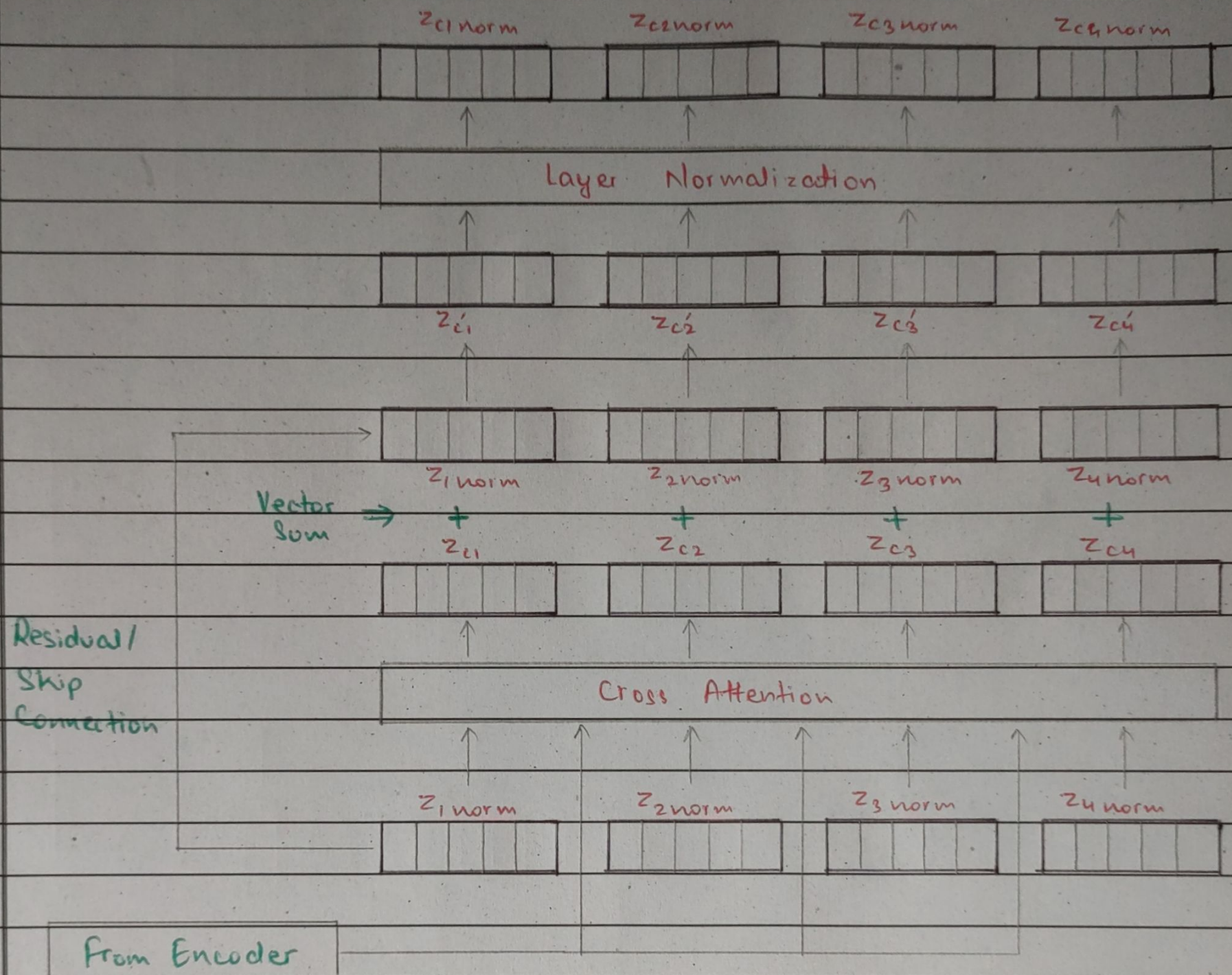
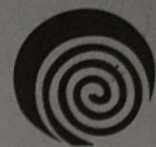




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- \*  $Z_1$  norm,  $Z_2$  norm,  $Z_3$  norm &  $Z_4$  norm are passed in Cross-Attention where they are used to construct Query vectors, along with encoded vectors from Encoder which also are passed in Cross-Attention which are used to construct Key & Value vectors. These are then used to calculate contextual embeddings of the Hindi text words provided as input to the Decoder,  $Z_{c1}$ ,  $Z_{c2}$ ,  $Z_{c3}$  &  $Z_{c4}$ .



- \* Through residual/skip connection, original  $z_{1\text{norm}}$ ,  $z_{2\text{norm}}$ ,  $z_{3\text{norm}}$  &  $z_{4\text{norm}}$  are added to  $z_{c1}$ ,  $z_{c2}$ ,  $z_{c3}$  &  $z_{c4}$  resulting in  $z'_{c1}$ ,  $z'_{c2}$ ,  $z'_{c3}$  &  $z'_{c4}$ .
- \*  $z'_{c1}$ ,  $z'_{c2}$ ,  $z'_{c3}$  &  $z'_{c4}$  are then passed through layer normalization to normalize embedding values, resulting in  $z_{c1\text{norm}}$ ,  $z_{c2\text{norm}}$ ,  $z_{c3\text{norm}}$  &  $z_{c4\text{norm}}$ .
- \* Dimensions of all the vectors in this stage is 512.